

MORPHOMETRIC ANALYSIS OF PEDDAVANKA BASIN, ANANTAPUR DISTRICT, ANDHRA PRADESH, INDIA

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ABSTRACT

Morphometric analysis and their relative parameters have been quantitatively carried out for the Peddavanka basin, Anantapur district, Andhra Pradesh, India. The quantitative analyses of the morphometric characteristics of the basin include Stream order, Stream length, Bifurcation ratio, Drainage density, Drainage frequency, Relief ratio, Elongation ratio and Circularity ratio. The foregoing analysis clearly indicates some relations among the various attributes of the morphometric aspects of the basin and helps to understand their role in sculpturing the surface of the region.

Introduction

The river Peddavanka rises from Mutchukota hills of Erramalai hill ranges and flows in southern direction and joins the Pennar River near Tadipatri, Anantapur district, Andhra Pradesh, India. The river originates in the southern part of Kurnool district and drains through northeastern part of Anantapur district covering an area of about 298km². The Peddavanka basin lies between latitudes 14°55'24" - 15°12'50" and longitudes 77°51'20" - 78°03'26". The relief of the basin is 323 mts with maximum and minimum heights ranging between 552 mts and 229 mts respectively.

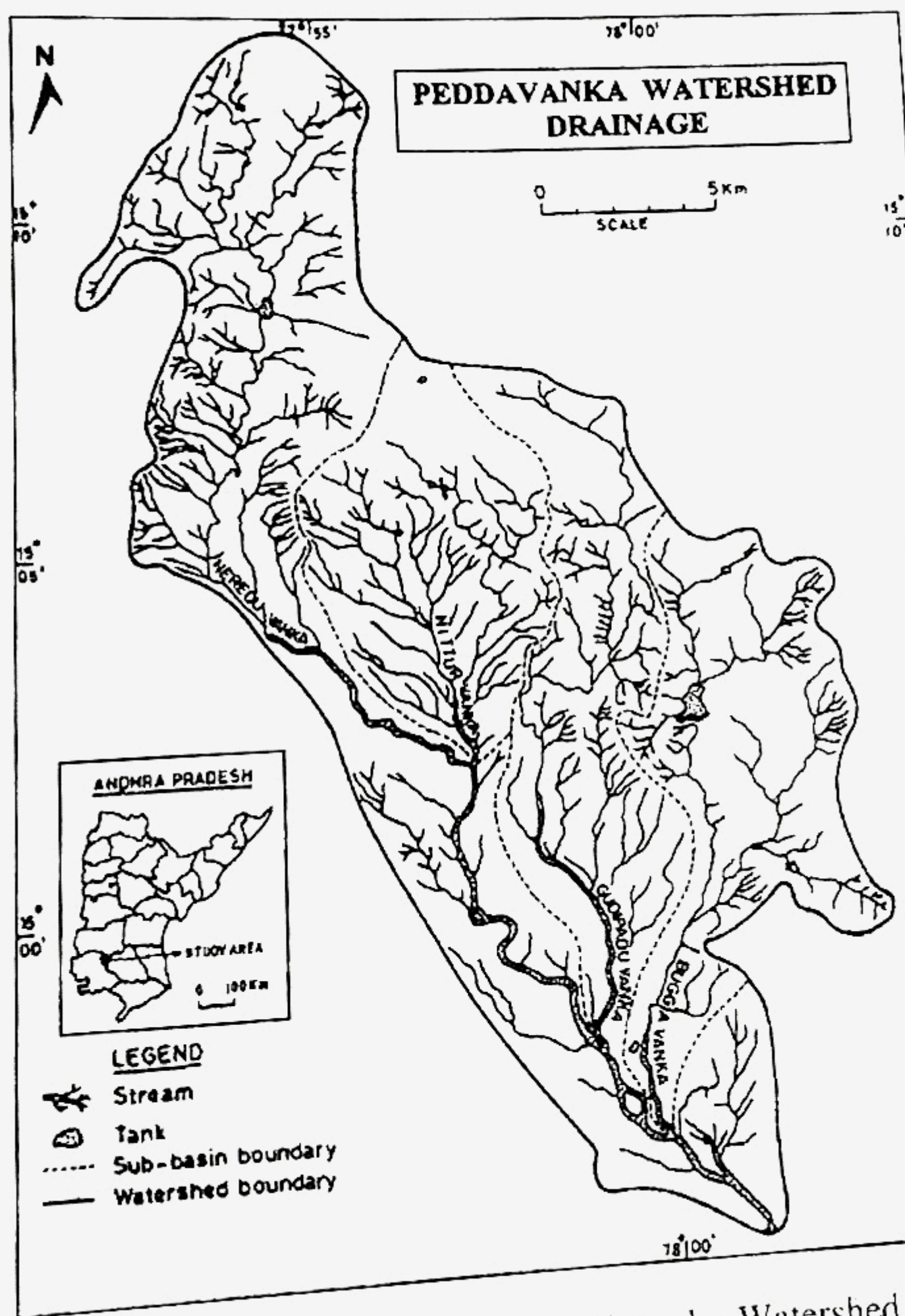
The major rock types in the area are Tadipatri shale and limestones of Cuddapah super group, Banaganapalle quartzite, Narji limestone, Auk shales and Paniam quartzite of Kurnool group. The climate of the basin is semi-arid in nature. It is characterised by general aridity but receives rain during the monsoon for a short period of the year. The basin receives an average annual rainfall of 550 mm out of which 90 percent in the two monsoon periods.

Analysis and Discussion

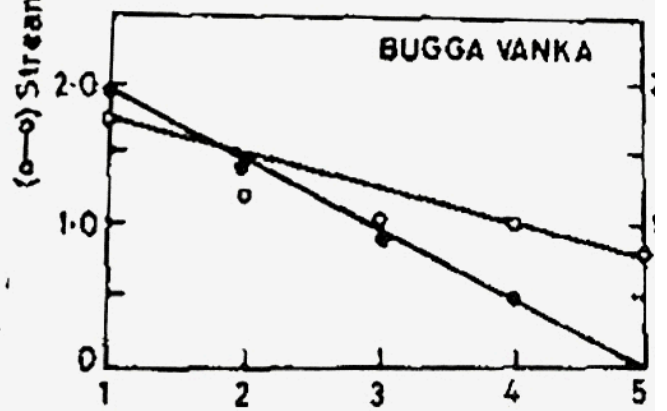
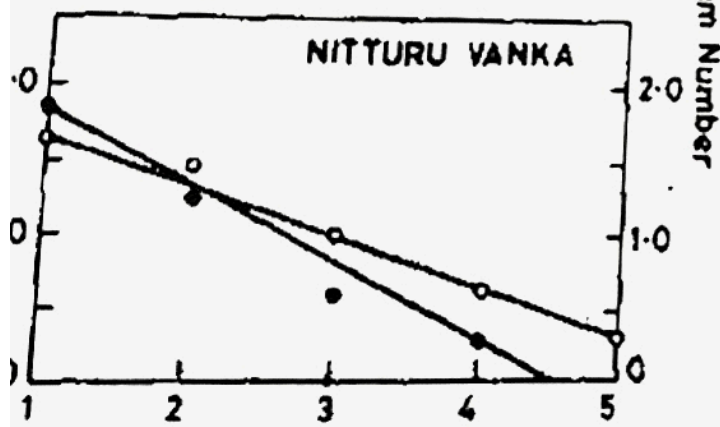
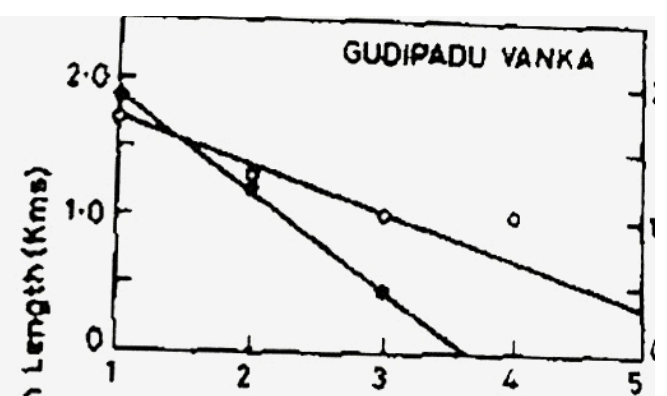
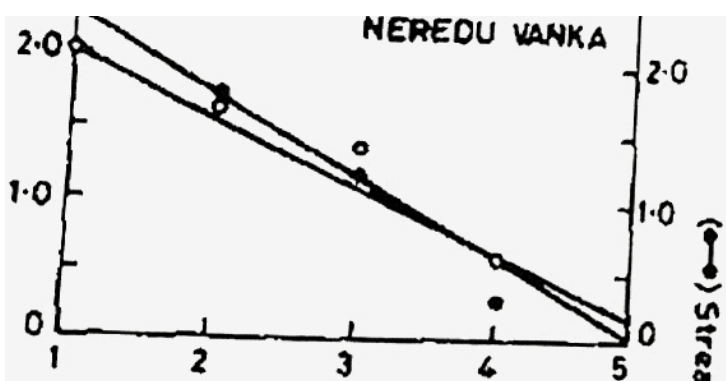
The study is primarily based on published and derived data. For the analysis of the drainage characteristics and relief, intensive use has been made of Survey of India Topographical sheets (1977) on 1: 50,000 scale. The quantitative analysis of the morphometric characteristics - Stream Order, Stream Length, Bifurcation Ratio, Drainage density, Stream frequency, Relief ratio, Elongation ratio and Circularity ratio-measured and calculated for Peddavanka basin are given in Tables - 1,2,3,4.

Table 1 : Stream orders and Stream lengths of Peddavanka basin

Basin / Sub-Basin	Stream numbers in different orders							Order wise total stream lengths (Km)							Stream length ratio				
	1	2	3	4	5	6	Total	1	2	3	4	5	6	Total	1/2	2/3	3/4	4/5	5/6
Neredi vanka	203	56	17	2	1		279	107.5	48.0	26.5	4.5	17.5		204.0	2.24	1.81	5.89	0.26	10.9
Nittur Vanka	76	18	4	2	1		101	48.0	28.0	10.5	4.5	2.0		93.0	1.71	2.67	2.34	2.25	
Godipadu Vanka	86	17	3	1			107	53.0	20.0	13.5	13.0			99.5	2.65	1.48	1.04		
Bugga Vanka	96	25	8	3	1		133	54.0	16.0	13.0	10.0	6.0		99.0	3.37	1.23	1.30	1.67	
Peddavanka Basin	461	116	32	8	3	1	621	262.5	112.0	63.5	32.0	25.5	16.0	511.5	2.34	1.76	1.98	1.25	1.29

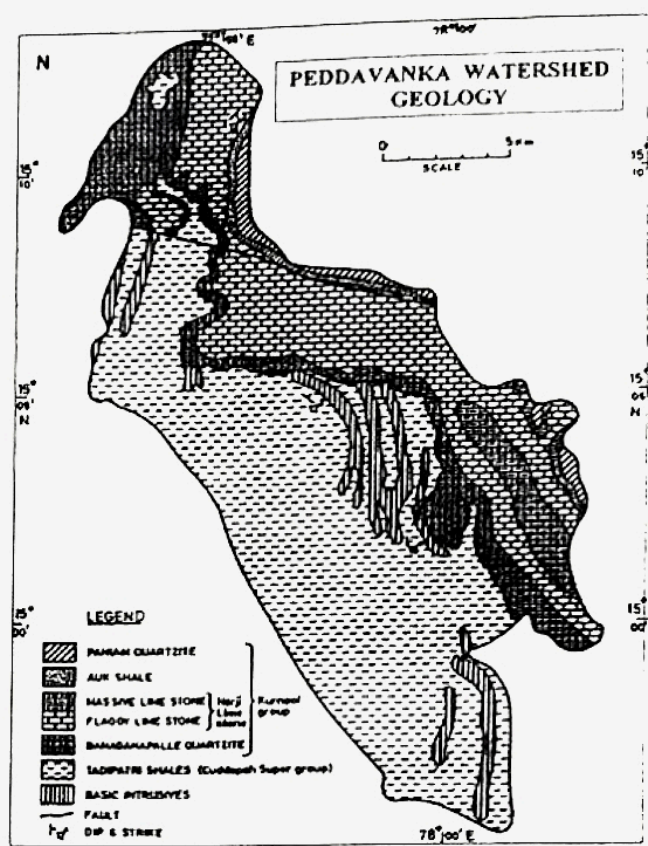
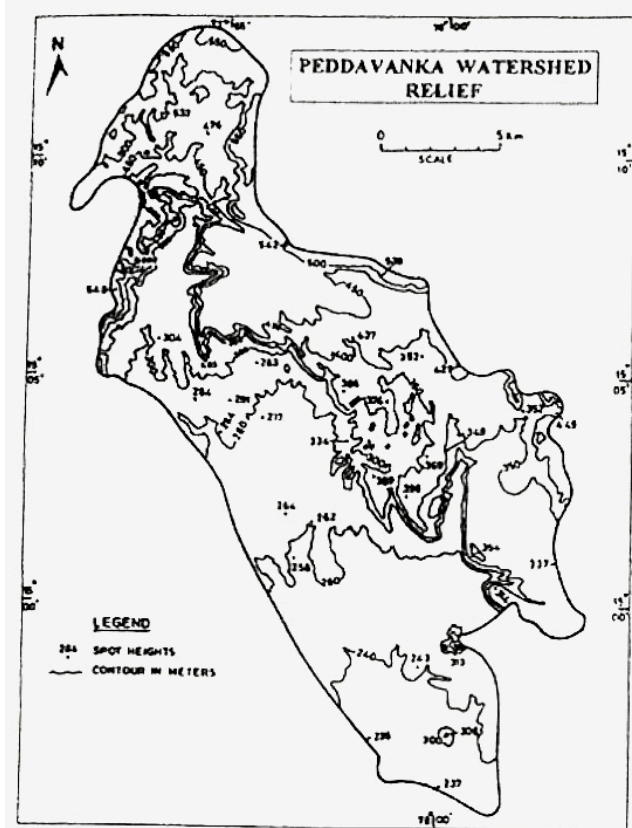


Peddavanka Watershed



Basin / Sub-basin	Area In Km ²	Drainage Frequency (Km/Km ²)	Length (in Km)	Form Factor	Elongation Ratio	Circulatory Ratio
Neredu Vanka	137	2.04	43.5	0.07	0.30	0.18
Nittur Vanka	50	2.02	13.2	0.29	0.60	0.71
Gudipadu Vanka	52	2.06	26.5	0.07	0.31	0.31
Bugga Vanka	59	2.25	22.0	0.12	0.39	0.32
Peddavanka Basin	298	2.08	43.5	0.16	0.45	0.36

Basin / Sub-basin	Drainage Density (Km/Km ²)	Drainage Texture	Bifurcation Ratios					
			Rb1	Rb2	Rb3	Rb4	Rb5	Mean Rb
Neredu Vanka	1.49	3.04	3.63	3.29	8.50	2.00		4.36
Nittur Vanka	1.86	3.76	4.22	4.50	2.00	2.00		3.18
Gudipadu Vanka	1.91	3.93	5.06	5.67	3.00			4.58
Bugga Vanka	1.68	3.78	3.84	3.13	2.67	3.00		3.16
Peddavanka Basin	1.72	3.58	3.97	3.63	4.00	2.67	3.00	3.45



The Peddavanka Basin is a 6th order stream with dendritic drainage pattern, which is a characteristic feature of regions occupied by mainly horizontal sedimentary rocks. The slope varies from '0' to 17°27' in the basin. High degree of slope is noticed in the northern, eastern and central parts of the basin, mostly occupied by the limestones. The shale formations are characterized with low slope areas showing less than 4°, in the southern part of the basin. From the plots of stream order with stream numbers and lengths, it is revealed that they are supporting the Horton's law of stream numbers and stream lengths. In almost all cases, the basin length decreases as the order increases and it is lowest in case of the highest stream orders. The stream length ratios are changing haphazardly both at the basin and sub-basin length, and it has an important relationship with the surface flow discharge and erosional stage of the basin. The elongation ratio of Peddavanka basin is 0.45, showing extremely elongated nature and variations of the elongated shape of the basins are due to the guiding effect of the thrusting and faulting. The drainage density of Peddavanka basin as well as the sub-basins reveals that the nature of sub-surface strata is permeable, which is a characteristic feature of coarse drainage density, since the value is less than 5.0. The drainage texture of the basin and sub-basins are coarse textured, as the values are less than 4.0. Bifurcation ratio varies from 2.67 to 4.00 for Peddavanka basin with mean bifurcation ratio of 3.45. The mean bifurcation ratios of all the sub-basins lie in between 3 and 5 indicating the homogeneous characters and geologic structure. It is concluded that higher values of bifurcation ratio indicate a mature topography, which is the result of the process of drainage integration. The relief of the basin is 323 mts and the relief ratio of the basin is 0.007 and from the value of elongation ratio (0.45) it is found that the study area possesses strong relief. In general the relief ratio of the basin and sub-basins are low which is characteristic of less resistant rocks of the area.

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